

JAVA PROGRAMMING LAB

Queue Interface Methods Demonstration

Experiment No.	05
Date	21/09/26
Title	Demonstration of Queue Interface Methods in Java

1. Objective / Aim

To demonstrate the commonly used methods of the Java Queue interface such as **add()**, **offer()**, **element()**, **peek()**, **remove()**, and **poll()** using a LinkedList implementation.

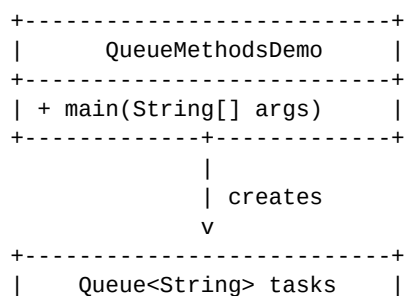
2. System Requirements

Hardware Requirements	Software Requirements
Computer/Laptop	Java Development Kit (JDK 17 or later)
Minimum 4 GB RAM	Java compiler (javac) and Java Runtime Environment (java)
At least 100 MB free storage	IDE: Eclipse / IntelliJ IDEA / NetBeans / VS Code (optional)
Keyboard and monitor	Operating System: Windows / Linux / macOS

3. Algorithm / Logic Description

1. Start the program.
2. Import the Queue interface and LinkedList class.
3. Create a Queue of String values using LinkedList.
4. Insert three tasks using add().
5. Insert another task using offer().
6. Display the first element using element() and peek().
7. Remove the head element using remove() and display the updated queue.
8. Remove the next head element using poll() and display the updated queue.
9. Stop the program.

4. UML Class Diagram



5. Source Code

```
Head using peek(): Study
Removed using remove(): Study
Queue after remove(): [Practice, Assignment, Project]
Removed using poll(): Practice
Queue after poll(): [Assignment, Project]
```

Result: The Queue interface methods add(), offer(), element(), peek(), remove(), and poll() were successfully demonstrated and executed using a LinkedList.